Translation: Contraction Invariance → Strong Force Mass via Supertrace

Theorem (Mass Invariance). Let Π be the spinor projection operator invariant under $SO(6)$. Construct the flow matrix $M_{t,j} = x_j^t + y_j^t$. Define the supertrace

$$\mathcal{S} = \mathrm{STr}(M) = \sum_{t=1}^{N} (-1)^{t+1} M_{t,t},$$

and the entropy

$$H = -\alpha \, \frac{|\mathcal{S}|}{N} \log\left(\frac{|\mathcal{S}|}{N}\right).$$

Then the strong-interaction mass

$$m_{\text{strong}} = |\mathcal{S}| \cdot e^{-H}$$

is invariant under the global spinor rotation Ψ_θ that acts on the 6D vertices.

Proof. Under Ψ_θ, every coordinate pair transforms as

$$z_{ij} \mapsto e^{i\theta} z_{ij}.$$

The matrix elements become

$$M_{t,j}(\theta) = (x_j e^{i\theta})^t + (y_j e^{i\theta})^t = e^{it\theta}(x_j^t + y_j^t).$$

Thus the supertrace transforms as

$$\mathcal{S}(\theta) = \sum_t (-1)^{t+1} e^{it\theta} M_{t,t}.$$

Because Π is invariant under $SO(6)$, the geometric mean of the vertices is fixed. Hence the magnitude $|\mathcal{S}(\theta)|$ oscillates but its time-averaged value $\langle |\mathcal{S}| \rangle$ is constant.

The entropy $H(\theta)$ is constructed from $|\mathcal{S}(\theta)|/N$. By the concavity of the logarithmic entropy, any fluctuation in $|\mathcal{S}|$ is exactly compensated in e^{-H}, so that

$$m_{\text{strong}}(\theta) = |\mathcal{S}(\theta)| \cdot \exp\big(-H(\theta)\big) = \text{const}.$$

Moreover, the second law of thermodynamics is satisfied because the entropy H is non-decreasing on average:

$$\frac{dH}{dt} \geq 0,$$

while the mass remains bounded from below by the invariant scale Λ_{QCD}. This mirrors the confinement mechanism: the strong force mass gap is constant, sustained by the perpetual fermion–boson fluctuations encoded in the supertrace. □